

Coefficientwise magic positivity through dimension ten

An exact-arithmetic partial solution for
generalized parking-function polytopes

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Abstract

Hill, Luo, Trinh, and Vindas-Meléndez conjecture that every generalized parking-function polytope $\mathfrak{X}_n(\mathbf{b})$ with $n \geq 3$ is magic positive: its Ehrhart polynomial has nonnegative coefficients in the basis $t^j(t+1)^{n-j}$. We obtain an exact-arithmetic, computer-assisted proof for *all positive integer parameter vectors* in every dimension $3 \leq n \leq 10$. The certified statement is stronger: after $b_i = 1 + x_i$, every coefficient of every polynomial $n!_{\mu_{n,j}}(\mathbf{1} + \mathbf{x})$ is nonnegative. For $4 \leq n \leq 10$, all monomials of total degree at most j occur with strictly positive coefficients. Dimension three has a short explicit proof printed in full. Independently of the dimension bound, we derive a harmonic-number formula for $\mu_{n,1}$ and prove its sharp minimum $n(3n - 7)/4$. We also identify the zero cases of the final magic coefficient and give a cube-addition identity reducing the general problem to $b_1 = 1$. The accompanying package contains exact polynomial certificates, a standard-library Python generator and verifier, and independent lattice-point checks. The unrestricted conjecture is not settled here; the distinction between the finite-dimensional theorem and the remaining all-dimensional problem is maintained throughout.

Keywords. Ehrhart polynomial; magic positivity; generalized parking function; polymatroid; exact polynomial certificate.

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1 The selected problem and the result obtained

1.1 Selection, exclusion, and current source status

The subject was selected before choosing the problem. A catalogue of 120 mathematical areas was fixed, and a pseudorandom draw seeded by 128 bits from the operating system selected entry 33, *discrete geometry*. There was no redraw. The seed, ordered catalogue, and replay command are recorded in Appendix A and in `data/selection.json`.

The supplied `manifest(1).tex` was read in full and used as an exclusion list. Its 71 entries do not contain the problem considered here. In particular, its lucky-parking-spots problem concerns a statistic on parking functions; its preorder-polytope and order/chain-polytope problems concern different polytopes and different invariants. The present question concerns the Ehrhart polynomial of the convex hull of generalized parking functions. The input-file hash and the exclusion comparison are recorded in `notes/exclusion_audit.md`.

The precise target is Conjecture 8.1 of Hill, Luo, Trinh, and Vindas-Meléndez [1]. Their preprint, arXiv:2607.15503v1, was submitted on 16 July 2026. That version explicitly leaves arbitrary parameter vectors open, while proving the two-parameter family $(a, b, \dots, b)$. The version history and targeted literature searches were checked on 20 September 2026; no later resolution of the arbitrary-vector conjecture was located. The investigation below uses that published formulation.

1.2 Definitions and the conjecture

For $\mathbf{b} = (b_1, \dots, b_n) \in \mathbb{Z}_{\geq 0}^n$, put

$$S_i = b_1 + \dots + b_i, \quad S_0 = 0.$$

A ***b*-parking function** is a positive integer vector whose increasing rearrangement $\beta'_1 \leq \dots \leq \beta'_n$ satisfies $\beta'_i \leq S_i$. Let $\mathfrak{X}_n(\mathbf{b})$ be the convex hull of these vectors, and define

$$E_{n,\mathbf{b}}(t) = \#(t\mathfrak{X}_n(\mathbf{b}) \cap \mathbb{Z}^n) \quad (t \in \mathbb{Z}_{\geq 0}).$$

The vertices are integral, and this counting function is a polynomial. We will also obtain its polynomiality directly from an exact recursion.

For $n \geq 2$, these polytopes have dimension n . Expand uniquely as

$$E_{n,\mathbf{b}}(t) = \sum_{j=0}^n \mu_{n,j}(\mathbf{b}) t^j (t+1)^{n-j}. \quad (1.1)$$

The word *magic* names the basis, not an additional geometric operation. The polytope is *magic positive* when all $\mu_{n,j}(\mathbf{b})$ are nonnegative. Equivalently, the polynomial

$$M_{n,\mathbf{b}}(y) = (1-y)^n E_{n,\mathbf{b}}\left(\frac{y}{1-y}\right) = \sum_{j=0}^n \mu_{n,j}(\mathbf{b}) y^j \quad (1.2)$$

has nonnegative ordinary coefficients.

Conjecture 1.1 (Hill–Luo–Trinh–Vindas-Meléndez). For every $n \geq 3$ and every $\mathbf{b} \in \mathbb{Z}_{\geq 0}^n$, the polytope $\mathfrak{X}_n(\mathbf{b})$ is magic positive.

Magic positivity is stronger than ordinary Ehrhart positivity. It also implies real-rootedness of the h^* -polynomial through Brändén’s transformation theorem [3, Theorem 4.2]. This explains why changing the basis is useful rather than merely cosmetic. Section 6 makes the implication precise.

1.3 Main results and their proof status

Theorem 1.2 (Coefficientwise positivity through dimension ten). *For $3 \leq n \leq 10$ and $0 \leq j \leq n$,*

$$n! \mu_{n,j}(1 + x_1, \dots, 1 + x_n) \in \mathbb{Z}_{\geq 0}[x_1, \dots, x_n]. \quad (1.3)$$

For $4 \leq n \leq 10$, every monomial of total degree at most j has a strictly positive coefficient. For $n = 3$, the same assertion holds except that the constant coefficient of $6\mu_{3,3}(\mathbf{1} + \mathbf{x})$ is zero. Consequently Conjecture 1.1 holds for all positive integer parameter vectors in these eight dimensions.

The dimension-three proof is entirely explicit in Section 4. The proof for dimensions four through ten is computer-assisted: exact polynomial identities obtained from a proved recursion are checked, and their complete coefficient lists are supplied. It is not a numerical test over a finite box of parameter values.

Theorem 1.3 (The first magic coefficient in every dimension). *Let $H_m = \sum_{k=1}^m 1/k$. For $n \geq 3$ and $x_i = b_i - 1 \geq 0$,*

$$\mu_{n,1}(\mathbf{1} + \mathbf{x}) = \frac{n(3n-7)}{4} + nx_1 + \sum_{r=1}^{n-1} r(1 + H_n - H_r)x_{n-r+1}. \quad (1.4)$$

In particular,

$$\mu_{n,1}(\mathbf{b}) \geq \frac{n(3n-7)}{4} > 0,$$

with equality precisely at $\mathbf{b} = \mathbf{1}$.

The second theorem has a proof in arbitrary dimension and does not rely on the certificates. The identities $\mu_{n,0} = 1$ and $\mu_{n,n} = \#(\text{int } \mathfrak{X}_n(\mathbf{b}) \cap \mathbb{Z}^n)$ further control the endpoints. Therefore any counterexample not ruled out by this work must have $n \geq 11$ and a negative coefficient at an index $2 \leq j \leq n-1$. This does *not* assert that such a counterexample exists.

The conjecture in all dimensions, and the coefficientwise strengthening of it, remain unproved in this draft.

2 Geometry and an exact counting recursion

The inequality description used here is due to Bayer and collaborators [2, Theorem 2.3]. The lattice-layer recursion, polynomiality, and dilation identity are established in [1, Theorem 3.5, Theorem 4.1, Proposition 5.4, and Section 5]. They are inputs rather than novelty claims. We provide direct derivations because the correctness of the certificates depends on their boundary conventions.

2.1 A symmetric rank description

Translate by the integer vector $\mathbf{1}$, and write

$$P_n(\mathbf{b}) = \mathfrak{X}_n(\mathbf{b}) - \mathbf{1}, \quad u_i = S_i - 1, \quad f(k) = \sum_{i=n-k+1}^n u_i, \quad f(0) = 0.$$

Then

$$P_n(\mathbf{b}) = \left\{ z \in \mathbb{R}_{\geq 0}^n : \sum_{i \in I} z_i \leq f(|I|) \text{ for every } I \subseteq [n] \right\}. \quad (2.1)$$

The sequence $u_1 \leq \dots \leq u_n$ is nonnegative. The same definition makes sense for the auxiliary weak parameter range

$$b_1 \geq 1, \quad b_2, \dots, b_n \geq 0. \quad (2.2)$$

We will need this range to evaluate the counting polynomial on coordinate axes. It is a genuine geometric extension, not an assumption that arbitrary polynomial specializations still count lattice points.

Lemma 2.1 (Greedy description). *The right side of (2.1) is the convex hull of all coordinate permutations of*

$$(u_n, u_{n-1}, \dots, u_{n-m+1}, 0, \dots, 0), \quad 0 \leq m \leq n.$$

This holds throughout (2.2). In particular the polytope is integral when the parameters are integers.

Proof. Each displayed vector satisfies all subset inequalities, since the sum of any k coordinates is at most the sum of the k largest u_i . For the reverse inclusion, maximize a linear functional. By symmetry, its positive coefficients may be ordered $c_1 \geq \dots \geq c_m > 0$; coordinates with nonpositive coefficients can be set to zero, since the feasible set is downward closed. With $c_{m+1} = 0$, every feasible z satisfies

$$\sum_{i=1}^m c_i z_i = \sum_{k=1}^m (c_k - c_{k+1}) \sum_{i=1}^k z_i \leq \sum_{k=1}^m (c_k - c_{k+1}) f(k).$$

Equality is attained by $z_i = u_{n-i+1}$ for $i \leq m$ and zero otherwise. Thus the proposed convex hull has the same support function. After adding **1**, the displayed vectors are parking functions, so this also identifies the original convex hull when the parameters are positive. $\square$

Remark 2.2. Counting lattice points of the convex hull is not the same as counting parking functions. For example, $(2, 2, 2) \in \mathfrak{X}_3(1, 1, 1)$, since its largest one-, two-, and three-coordinate sums are 2, 4, 6, bounded by 3, 5, 6. It is not a parking function: its smallest coordinate exceeds $S_1 = 1$. The algorithms below count points of the polytope, not just its defining integer set.

2.2 Derivation of the layers

In the unshifted coordinates put

$$\Sigma_k = \sum_{i=n-k+1}^n S_i.$$

Fix the last coordinate at an integer height h , where $1 \leq h \leq S_n$. For a k -element subset I of the other coordinates, the two relevant bounds are

$$\sum_{i \in I} x_i \leq \Sigma_k, \quad \sum_{i \in I} x_i \leq \Sigma_{k+1} - h.$$

Their minimum is the exact residual bound. Indeed all inequalities either contain the fixed coordinate or do not. The switch between the two bounds is controlled by

$$\Sigma_{k+1} - \Sigma_k = S_{n-k}.$$

Write $h = S_{\ell-1} + r$ with $1 \leq r \leq b_\ell$. The resulting layer is $\mathfrak{X}_{n-1}(\mathbf{b}^{(\ell, r)})$, where

$$\mathbf{b}^{(1, r)} = (b_1 + b_2, b_3, \dots, b_n), \quad (2.3)$$

$$\mathbf{b}^{(\ell, r)} = (b_1, \dots, b_{\ell-2}, b_{\ell-1} + b_\ell - r, b_{\ell+1} + r, b_{\ell+2}, \dots, b_n), \quad 2 \leq \ell \leq n-1, \quad (2.4)$$

$$\mathbf{b}^{(n, r)} = (b_1, \dots, b_{n-2}, b_{n-1} + b_n - r). \quad (2.5)$$

To check the indexing, the new partial sums for $\ell > 1$ are S_i for $i \leq \ell - 2$, then $S_\ell - r$, and then S_{i+1} for $i \geq \ell$; for $\ell = 1$ they are simply S_{i+1} . Their largest-coordinate sums give precisely $\min(\Sigma_k, \Sigma_{k+1} - h)$.

This derivation also covers zero values among $b_2, \dots, b_n$: a block of length zero contributes no layers. Every parameter vector that does occur still satisfies (2.2).

Proposition 2.3 (Exact lattice-count recurrence). *Let $L_n(\mathbf{b}) = \#(\mathfrak{X}_n(\mathbf{b}) \cap \mathbb{Z}^n)$ in the weak integer parameter range. Then $L_1(b_1) = b_1$, and for $n \geq 2$,*

$$L_n(\mathbf{b}) = b_1 L_{n-1}(b_1 + b_2, b_3, \dots, b_n) + \sum_{\ell=2}^{n-1} \sum_{r=1}^{b_\ell} L_{n-1}(\mathbf{b}^{(\ell,r)}) + \sum_{r=1}^{b_n} L_{n-1}(\mathbf{b}^{(n,r)}). \quad (2.6)$$

The middle sum is empty for $n = 2$.

Proof. The layers partition the lattice points. In the first block all b_1 layers have the same parameter vector (2.3); the other blocks use (2.4) and (2.5) individually. $\square$

2.3 Polynomiality, degree, and denominators

Proposition 2.4. *There is a unique polynomial $L_n \in \mathbb{Q}[b_1, \dots, b_n]$ of total degree at most n that gives these counts. Moreover,*

$$n! L_n \in \mathbb{Z}[b_1, \dots, b_n]. \quad (2.7)$$

Proof. Induct on n . Affine substitution preserves the degree bound $n - 1$. Expanding a summand in powers of r and summing r^k from 1 to b_ℓ replaces it by a polynomial of degree $k + 1$; hence the total degree rises by at most one. The first block also raises degree by at most one. This constructs L_n . Uniqueness follows because a polynomial vanishing on all positive integer vectors vanishes identically, by induction on the number of variables.

For the denominator statement, let $R(\mathbf{x}) = L_n(\mathbf{1} + \mathbf{x})$. Its values on $\mathbb{Z}_{\geq 0}^n$ are integers. The multivariate Newton expansion is

$$R(\mathbf{x}) = \sum_{|\alpha| \leq n} (\Delta_1^{\alpha_1} \cdots \Delta_n^{\alpha_n} R)(\mathbf{0}) \prod_{i=1}^n \binom{x_i}{\alpha_i}.$$

All finite differences are integers. Each denominator $\prod_i \alpha_i!$ divides $n!$, because $|\alpha| \leq n$. Multiplication by $n!$ therefore gives integral ordinary coefficients; translation back to the b_i preserves integrality. $\square$

The recurrence and the polynomial agree on the weak parameter domain, not merely on its strictly positive part. This follows from the layer proof, where zero-length blocks are omitted. This point is useful in the next section and in the all-dimensional coefficient calculation.

3 The magic transform and finite symbolic certificates

3.1 Dilation and homogeneous components

By the rank description, multiplication of $P_n(\mathbf{b})$ by a nonnegative integer t replaces u_i by tu_i . Consequently,

$$E_{n,\mathbf{b}}(t) = L_n(1 + t(b_1 - 1), tb_2, \dots, tb_n). \quad (3.1)$$

The translation between $t\mathfrak{X}_n(\mathbf{b})$ and $tP_n(\mathbf{b})$ is integral. At $t = 0$, the right side is $L_n(1, 0, \dots, 0) = 1$, as required. Thus (3.1) supplies an Ehrhart polynomial without an interpolation step.

Define

$$F_n(\mathbf{q}) = L_n(1 + q_1, q_2, \dots, q_n) = \sum_{d=0}^n F_{n,d}(\mathbf{q}),$$

where $F_{n,d}$ is homogeneous of degree d . If $E_{n,\mathbf{b}}(t) = \sum_{d=0}^n e_{n,d}(\mathbf{b})t^d$, then

$$e_{n,d}(\mathbf{b}) = F_{n,d}(b_1 - 1, b_2, \dots, b_n), \quad \mu_{n,j} = \sum_{d=0}^j (-1)^{j-d} \binom{n-d}{j-d} e_{n,d}. \quad (3.2)$$

In particular $\mu_{n,j}(\mathbf{1} + \mathbf{x})$ has degree at most j . The integer scaling in (2.7) survives all translations and the integral triangular transform.

There are two different translations here. The homogeneous grading uses $(q_1, q_2, \dots) = (b_1 - 1, b_2, \dots)$, whereas the positivity certificate uses $x_i = b_i - 1$ for *every* coordinate. Confusing these shifts would produce an incorrect certificate; the implementation performs them in separate stages.

3.2 A small exact summation kernel

For nonnegative integers a, b , introduce

$$K_{a,b}(X, Y, Z) = \sum_{r=1}^Y (X + Y - r)^a (Z + r)^b, \quad (3.3)$$

where the sum is interpreted as its unique polynomial continuation in Y . This kernel carries every middle block of the recursion. The final block uses $K_{a,0}(X, Y, 0)$.

For a complete construction, let $A_k(Y) = \sum_{r=1}^Y r^k$. Telescoping $(r+1)^{k+1} - r^{k+1}$ gives

$$A_k(Y) = \frac{(Y+1)^{k+1} - 1 - \sum_{h=0}^{k-1} \binom{k+1}{h} A_h(Y)}{k+1}. \quad (3.4)$$

After binomial expansion,

$$K_{a,b}(X, Y, Z) = \sum_{i+j+h=a} \binom{a}{i, j, h} X^i Y^j (-1)^h \sum_{z=0}^b \binom{b}{z} Z^z A_{h+b-z}(Y). \quad (3.5)$$

All arithmetic is rational, and $n!K_{a,b}$ has integral coefficients when $a + b + 1 \leq n$. This also follows from the integer-valued-polynomial argument of Proposition 2.4.

An independent identity is especially useful for auditing the kernel:

$$K_{a,b}(X, 0, Z) = 0, \quad (3.6)$$

$$K_{a,b}(X, Y, Z) - K_{a,b}(X + 1, Y - 1, Z) = X^a (Z + Y)^b. \quad (3.7)$$

The second term has exactly the first $Y - 1$ summands of the first; the remaining summand is the displayed right side. Conversely these identities uniquely determine the kernel on nonnegative integer Y , and therefore as a polynomial. The verifier checks them by independent polynomial translation, not by reapplying the power-sum recurrence.

3.3 What the certificate proves

A sparse polynomial is represented by a map

$$(\alpha_1, \dots, \alpha_n) \mapsto c_\alpha \in \mathbb{Z}.$$

The lattice files store $n!L_n(\mathbf{b})$, and the magic files store $n!\mu_{n,j}(\mathbf{1} + \mathbf{x})$. Missing monomials have coefficient zero. The three invariants of the generator are:

- (i) the current lattice map is exactly $n!L_n$, by the layer recursion;
- (ii) homogeneous extraction and (3.2) produce exactly the scaled magic maps;
- (iii) a nonnegative map evaluates nonnegatively whenever $x_i \geq 0$.

No root approximation, positivity tolerance, guessed recurrence, or parameter-bound assumption enters any invariant.

Computer-assisted proof of Theorem 1.2. Start from $L_1(b_1) = b_1$. Equations (2.6), (3.4), and (3.5) construct L_n inductively. Equations (3.1) and (3.2) construct the magic maps. The supplied verifier regenerates and compares every stored coefficient for $1 \leq n \leq 10$. For $3 \leq n \leq 10$, every retained coefficient is a positive integer. The number of retained monomials in the j th map is

$$\binom{n+j}{j} - \begin{cases} 1, & (n, j) = (3, 3), \\ 0, & \text{otherwise.} \end{cases} \quad (3.8)$$

The degree bound is j , so for $n \geq 4$ this is the entire possible support. For $n = 3, j = 3$, the single missing monomial is the constant, as the explicit formula below confirms. Evaluating at $x_i = b_i - 1 \geq 0$ proves the stated infinite-parameter assertions. The exact output is summarized in Table 1; the full coefficient lists, rather than only their counts, are included in the archive. $\square$

n	Terms in $n!L_n$	Terms in all magic maps	$\mu_{n,n}(\mathbf{1})$
3	16	34	0
4	60	126	5
5	227	462	96
6	851	1,716	1,758
7	3,223	6,435	34,820
8	12,200	24,310	762,675
9	46,456	92,378	18,489,870
10	177,525	352,716	493,794,476

Table 1: Exact certificate sizes and the final magic coefficient at the classical parameter vector. All 478,177 retained coefficients in the magic maps for $3 \leq n \leq 10$ are positive. The last column is also an interior lattice-point count.

The number of possible monomials in a polynomial of degree at most j in n variables is $\binom{n+j}{j}$. Summing over j gives $\binom{2n+1}{n}$, which explains the total support size for $n \geq 4$. These support counts are observations proved for the listed dimensions, not an all-dimensional extrapolation.

4 A complete explicit proof in dimension three

The first two recursion steps give

$$\begin{aligned} L_2(b_1, b_2) &= b_1^2 + 2b_1b_2 + \frac{1}{2}b_2^2 - \frac{1}{2}b_2, \\ L_3(b_1, b_2, b_3) &= b_1^3 + 3b_1^2b_2 + 3b_1b_2^2 + 3b_1b_2b_3 + 6b_1b_2b_3 + \frac{3}{2}b_1b_3^2 - \frac{3}{2}b_1b_3 \\ &\quad + \frac{5}{6}b_2^3 + 2b_2^2b_3 - \frac{1}{2}b_2^2 + b_2b_3^2 - 2b_2b_3 - \frac{1}{3}b_2 + \frac{1}{6}b_3^3 - \frac{1}{2}b_3^2 + \frac{1}{3}b_3. \end{aligned}$$

The L_2 formula follows by summing the last block of (2.6); substitution of it into the same recurrence gives L_3 using sums of first and second powers. Thus the following identities can be checked using only elementary polynomial arithmetic.

Put $x = b_1 - 1$, $y = b_2 - 1$, and $z = b_3 - 1$. Applying (3.2) gives

$$\begin{aligned} 6\mu_{3,0} &= 6, \\ 6\mu_{3,1} &= 18x + 16y + 11z + 9, \end{aligned} \tag{4.1}$$

$$6\mu_{3,2} = 18x^2 + x(36y + 27z + 27) + 15y^2 + 24yz + 22y + 6z^2 + 14z + 9, \tag{4.2}$$

$$\begin{aligned} 6\mu_{3,3} &= 6x^3 + 18x^2(y + z + 1) \\ &\quad + x(18y^2 + 36yz + 36y + 9z^2 + 27z + 18) \\ &\quad + 5y^3 + 12y^2z + 12y^2 + 6yz^2 + 12yz + 7y + z^3 + 3z^2 + 2z. \end{aligned} \tag{4.3}$$

All coefficients are nonnegative, and $x, y, z \geq 0$ for the desired parameter range. This proves Conjecture 1.1 in dimension three without any unprinted computation. Moreover, $\mu_{3,1}$ and $\mu_{3,2}$ are always positive, while $\mu_{3,3} = 0$ holds exactly when $x = y = z = 0$.

Example 4.1 (An asymmetric parameter vector). For $\mathbf{b} = (2, 3, 1)$, the formulas give

$$\begin{aligned} M_{3,\mathbf{b}}(y) &= 1 + \frac{59}{6}y + \frac{115}{3}y^2 + 54y^3, \\ E_{3,\mathbf{b}}(t) &= 1 + \frac{77}{6}t + 61t^2 + \frac{619}{6}t^3, \\ h_{3,\mathbf{b}}^*(z) &= 1 + 174z + 390z^2 + 54z^3. \end{aligned}$$

In particular there are 178 lattice points at dilation one. The first line agrees with the independently printed example in [1, Section 8]; that example was used as a check, not to fit the polynomial.

4.1 Why the lower-dimensional exception is genuine

For $u = b_1 - 1$ and $v = b_2 - 1$, the degree-two magic coefficients are

$$2\mu_{2,0} = 2, \quad 2\mu_{2,1} = 4u + 3v - 1, \quad 2\mu_{2,2} = 2u^2 + 4uv + v(v - 1).$$

For nonnegative integers u, v , the final expression is nonnegative; the middle one is negative precisely when $u = v = 0$. This rederives the known exception $\mathfrak{X}_2(1, 1)$ and shows why a proof that assumes coefficientwise positivity beginning in dimension two would fail. The nonnegative-value statement and the nonnegative-ordinary-coefficient statement are distinct: $v(v - 1)$ has a negative linear coefficient but is nonnegative at every nonnegative integer.

In dimension one, $\mathfrak{X}_1(b_1) = [1, b_1]$ has Ehrhart polynomial $1 + (b_1 - 1)t$. For $b_1 \geq 2$ its degree-one magic transform is $1 + (b_1 - 2)y$. The case $b_1 = 1$ is a point of intrinsic dimension zero; its intrinsic magic polynomial is 1. The auxiliary certificate files use ambient degree one in that case, yielding $1 - y$; this is not a counterexample to the intrinsic-dimensional definition.

5 A sharp all-dimensional formula for the first coefficient

5.1 Linearity from the multivariate counting polynomial

Equation (3.1) shows that $e_{n,1}(\mathbf{b})$ is the homogeneous linear part of $F_n(\mathbf{q})$ evaluated at $\mathbf{q} = (b_1 - 1, b_2, \dots, b_n)$. Hence it is enough to evaluate that linear form on the n coordinate axes. This observation avoids invoking a general mixed-Ehrhart theorem or assuming a positivity property of mixed volumes.

For $1 \leq r \leq n$, define the uniform-rank polytope

$$Q_{r,n} = \{z \in [0, 1]^n : z_1 + \dots + z_n \leq r\}. \quad (5.1)$$

On the q_1 axis, the rank polytope is a cube $q_1 Q_{n,n}$. On the q_j axis for $j \geq 2$, its u -sequence has $j - 1$ zeros followed by $n - j + 1$ copies of q_j , so it is $q_j Q_{n-j+1,n}$. These statements follow directly from the subset inequalities, including their weak-parameter interpretation.

5.2 The linear Ehrhart coefficient of a uniform-rank polytope

Lemma 5.1. *For $1 \leq r \leq n$,*

$$[t]E_{Q_{r,n}}(t) = r(1 + H_n - H_r). \quad (5.2)$$

Proof. At dilation t , count integer vectors $0 \leq z_i \leq t$ with $\sum_i z_i \leq rt$. Inclusion-exclusion for the upper coordinate bounds and the stars-and-bars formula give the polynomial identity

$$E_{Q_{r,n}}(t) = \sum_{j=0}^{r-1} (-1)^j \binom{n}{j} \binom{(r-j)t + n - j}{n}. \quad (5.3)$$

For positive integer t , intersections of $j \geq r$ forbidden events are empty, so they are not included. The terms shown with an insufficient nonnegative upper binomial argument vanish in the usual combinatorial way. Consequently (5.3) is valid for every positive integer t and is a polynomial identity.

The derivative of the $j = 0$ term at zero is rH_n , since

$$\binom{rt + n}{n} = \prod_{k=1}^n \left(1 + \frac{rt}{k}\right).$$

For $1 \leq j \leq r - 1$, the binomial polynomial has one zero factor at $t = 0$. Differentiating that factor gives

$$\left. \frac{d}{dt} \binom{(r-j)t + n - j}{n} \right|_{t=0} = \frac{(r-j)(n-j)!(-1)^{j-1}(j-1)!}{n!}.$$

After multiplication by $(-1)^j \binom{n}{j}$, its contribution is $-(r-j)/j$. Therefore

$$[t]E_{Q_{r,n}} = rH_n - \sum_{j=1}^{r-1} \frac{r-j}{j} = r(1 + H_n - H_r).$$

The formula includes $r = n$, where $Q_{n,n}$ is the unit cube and the answer is n . □

Proof of Theorem 1.3. Coordinate-axis evaluation of the linear form gives

$$e_{n,1}(\mathbf{b}) = n(b_1 - 1) + \sum_{r=1}^{n-1} r(1 + H_n - H_r)b_{n-r+1}. \quad (5.4)$$

Since $e_{n,0} = 1$, the triangular transform gives $\mu_{n,1} = e_{n,1} - n$. Substitute $b_i = 1 + x_i$ and evaluate the constant:

$$\begin{aligned} \sum_{r=1}^{n-1} r(1 + H_n - H_r) &= \frac{n(n-1)}{2} + \sum_{k=2}^n \frac{1}{k} \sum_{r=1}^{k-1} r \\ &= \frac{n(n-1)}{2} + \frac{n(n-1)}{4} = \frac{3n(n-1)}{4}. \end{aligned}$$

This proves (1.4). Every coefficient of an x_i is strictly positive. Thus the minimum on $x_i \geq 0$ is attained uniquely at $\mathbf{x} = \mathbf{0}$, where it equals $n(3n-7)/4$. $\square$

Remark 5.2. At the extremes, the coefficients of b_2 and b_n in (5.4) are $(n^2-1)/n$ and H_n , respectively. For $n = 3$, these are $8/3$ and $11/6$, reproducing the asymmetric weights in (4.1). At $n = 2$, the constant minimum becomes $-1/2$, exactly the exceptional magic coefficient above.

6 Endpoints, consequences, and parameter monotonicity

6.1 The first and last entries of the magic vector

Proposition 6.1. *For all positive parameters and $n \geq 3$,*

$$\mu_{n,0}(\mathbf{b}) = 1, \quad \mu_{n,n}(\mathbf{b}) = \#(\text{int } \mathfrak{X}_n(\mathbf{b}) \cap \mathbb{Z}^n).$$

Moreover $\mu_{n,n}(\mathbf{b}) = 0$ if and only if $n = 3$ and $\mathbf{b} = (1, 1, 1)$.

Proof. The constant coefficient follows from $E(0) = 1$. At $t = -1$, every term in (1.1) vanishes except the last, so $\mu_{n,n} = (-1)^n E(-1)$. Ehrhart–Macdonald reciprocity identifies this with the interior lattice count; this standard theorem is an external input [4, 5].

At $\mathbf{b} = \mathbf{1}$, the original-coordinate upper bound for a subset of size k is

$$\Sigma_k = \frac{k(2n - k + 1)}{2}.$$

For $n \geq 4$ this is strictly greater than $2k$ for every $1 \leq k \leq n$. Thus $(2, \dots, 2)$ strictly satisfies every lower and upper inequality and is an interior lattice point. Increasing any parameter preserves these strict inequalities.

For $n = 3$ at $\mathbf{b} = \mathbf{1}$, every interior lattice point would have all coordinates at least two and total sum strictly less than six, which is impossible. If any parameter is increased, the total-sum upper bound becomes greater than six, while the bounds on one- and two-coordinate sums were already strictly greater than two and four. Hence $(2, 2, 2)$ becomes interior. This proves the exact zero classification. $\square$

6.2 Consequences of the finite-dimensional theorem

For $3 \leq n \leq 10$, the coefficientwise assertion has implications stronger than nonnegativity at integer parameters. The polynomial continuations $\mu_{n,j}(\mathbf{1} + \mathbf{x})$ are coordinatewise nondecreasing for real $x_i \geq 0$, and for $j \geq 1$ they are strictly increasing in each coordinate. Their minimum is therefore attained at the classical vector $\mathbf{b} = \mathbf{1}$. For $4 \leq n \leq 10$, every mixed partial derivative of total order at most j is strictly positive on this orthant; the full-support statement proves this by differentiating its matching monomial. These are statements about the polynomial continuations, not an assertion that an irrational polytope has an Ehrhart polynomial.

Since $\mu_{n,0} = 1$, expanding the nonnegative magic basis gives

$$[t^k]E_{n,\mathbf{b}}(t) \geq \binom{n}{k} > 0 \quad (0 \leq k \leq n, \ 3 \leq n \leq 10). \quad (6.1)$$

The comparison is with $(1+t)^n$ coefficient by coefficient.

Define the h^* -polynomial by

$$\sum_{t \geq 0} E_{n,\mathbf{b}}(t) z^t = \frac{h_{n,\mathbf{b}}^*(z)}{(1-z)^{n+1}}.$$

Let $\mathcal{E}$ be the linear map sending $\binom{t}{k}$ to w^k . The identity $\sum_{t \geq 0} \binom{t}{k} z^t = z^k / (1-z)^{k+1}$ gives

$$h_{n,\mathbf{b}}^*(z) = (1-z)^n \mathcal{E}(E_{n,\mathbf{b}}) \left(\frac{z}{1-z} \right).$$

Brändén's theorem says that nonnegative magic coefficients force $\mathcal{E}(E)$ to have all its zeros in $[-1, 0]$ [3, Theorem 4.2 and equation (6.4)]. The fractional-linear substitution takes these zeros to nonpositive real zeros; a zero at -1 accounts for a possible degree drop. Thus the finite-dimensional theorem also gives real-rooted h^* -polynomials for every positive parameter vector with $3 \leq n \leq 10$. Their coefficient sequences are consequently log-concave and unimodal. The real-rootedness implication is classical; only its application to the newly certified parameter range is part of the present result.

7 Cube addition and a reduction to the boundary parameter

The following identity clarifies which parameter is easiest to remove from a possible all-dimensional proof. It is an exact identity, not a positivity assumption about general Minkowski sums.

Lemma 7.1 (Adding coordinate segments). *Let $P \subset \mathbb{R}_{\geq 0}^n$ be a compact downward-closed convex set. For an integer $m \geq 0$ and a coordinate vector e_i ,*

$$\#((P + [0, m]e_i) \cap \mathbb{Z}^n) = \#(P \cap \mathbb{Z}^n) + m \#(\text{proj}_{[n] \setminus \{i\}} P \cap \mathbb{Z}^{n-1}).$$

Proof. Over an integer point of the projection, the original fiber is a nonempty interval $[0, u]$. It contains $\lfloor u \rfloor + 1$ integers. After adding the segment it is $[0, u + m]$, containing exactly m more. No other fibers appear. Downward closure is what guarantees that a nonempty projected fiber contains the integer height zero. $\square$

By the greedy support description,

$$P_n(b_1 + a, b_2, \dots, b_n) = P_n(\mathbf{b}) + a[0, 1]^n \quad (a \geq 0). \quad (7.1)$$

For $1 \leq k \leq n$, a projection onto any k coordinates has the same rank bounds through size k . Its parameter vector is

$$\pi_k \mathbf{b} = (S_{n-k+1}, b_{n-k+2}, \dots, b_n).$$

Put $E_{0,\pi_0 \mathbf{b}} = M_{0,\pi_0 \mathbf{b}} = 1$ as an empty-dimensional convention.

Proposition 7.2 (Cube-addition formula). *For nonnegative integer a and positive $\mathbf{b}$,*

$$E_{n,(b_1+a,b_2,\dots,b_n)}(t) = \sum_{k=0}^n \binom{n}{k} (at)^{n-k} E_{k,\pi_k \mathbf{b}}(t), \quad (7.2)$$

$$M_{n,(b_1+a,b_2,\dots,b_n)}(y) = \sum_{k=0}^n \binom{n}{k} (ay)^{n-k} M_{k,\pi_k \mathbf{b}}(y). \quad (7.3)$$

The degree- k ambient transform is used on the right; the projected one-dimensional parameters occurring for $k < n$ are at least two.

Proof. At an integer dilation t , apply Lemma 7.1 successively in all coordinates, with segment length at . Choosing the coordinates in which the projection term is taken produces the sum over all subsets. Symmetry identifies all terms with the same remaining dimension k , giving (7.2). Projection commutes with addition of the other coordinate segments, so no mixed term is omitted.

Both sides are polynomials in t . Substituting $t = y/(1 - y)$ and multiplying by $(1 - y)^n$ turns each term into $\binom{n}{k}(ay)^{n-k}M_{k,\pi_k\mathbf{b}}(y)$, which proves (7.3). $\square$

Corollary 7.3 (Boundary reduction). *To prove Conjecture 1.1 in every dimension, it is sufficient to prove it for the parameter vectors $(1, b_2, \dots, b_n)$ in every dimension $n \geq 3$.*

Proof. Use induction on n , starting with the elementary classifications in dimensions one and two. For a desired vector, apply (7.3) with base vector $(1, b_2, \dots, b_n)$ and $a = b_1 - 1$. The term $k = n$ is nonnegative by the boundary hypothesis. Terms $3 \leq k < n$ are nonnegative by induction. For $k = 2 < n$, the first projected parameter is at least $n - 1 \geq 2$, excluding the only non-magic-positive two-dimensional vector $(1, 1)$. For $k = 1 < n$, it is at least $n \geq 3$, so its magic transform is nonnegative. The term $k = 0$ is nonnegative as well. $\square$

This is a reduction of the difficulty, not its disappearance: the boundary family still has $n - 1$ independent positive parameters. Nor does the formula say that arbitrary Minkowski addition preserves magic positivity. Its special form depends on coordinate segments and downward-closed fibers.

8 Verification, reproducibility, and the limits of the attempt

8.1 Independent checks performed

The recorded run of `code/verify_certificates.py` completed with status `PASS`. It performs full coefficient-by-coefficient regeneration of all lattice and magic certificates through dimension ten, checks the degree and support claims, and compares every first-coefficient map with the independent harmonic-number formula. It also verifies all 55 kernel difference identities with $a + b \leq 9$ by direct polynomial translation.

The independent geometric enumerator uses only (2.1). It lists decreasing nonnegative integer tuples $z_1 \geq \dots \geq z_n$ and tests

$$z_1 + \dots + z_k \leq tf(k) \quad (1 \leq k \leq n).$$

These are equivalent to every subset inequality, since the largest sum of k coordinates is the prefix sum. A tuple with multiplicities $m_1, m_2, \dots$ represents exactly $n! / \prod_i m_i!$ coordinate permutations. Summing these orbit sizes gives the full lattice count. This procedure uses neither the slice recurrence nor power sums nor Ehrhart interpolation.

n	Parameter vectors	Dilations	Counts
3	$\{1, 2, 3, 4\}^3$	0, 1, 2, 3	256
4	$\{1, 2, 3\}^4$	0, 1, 2	243
5	$\{1, 2\}^5$	0, 1, 2	96
6	$\{1, 2\}^6$	0, 1	128
7–10	$\mathbf{b} = \mathbf{1}$	0, 1	8
Total			731

There are 486 positive-dilation checks and 245 zero-dilation checks. Each agrees both with the lattice polynomial under (3.1) and with the magic-basis expansion. All inputs and resulting integer counts are recorded in `data/verification.json`. These finite geometric checks are error-detection tests; the all-parameter conclusion comes from the symbolic identities and nonnegative coefficients, not from extrapolating the tests.

8.2 Reproducing and using the package

The core programs use only the Python 3.9+ standard library. From the unpacked package directory:

```
python code/verify_certificates.py
python code/evaluate_polytope.py 2 3 1
python code/sparse_magic.py --max-dimension 10 --output data/rebuilt
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command regenerates the full certificates before checking them. The evaluator returns exact rational Ehrhart and magic coefficients, integer h^* -coefficients, and a requested lattice count. It reads the provided certificates and accepts dimensions two through ten. Higher symbolic dimensions can be attempted with the generator, but are not part of the theorem certified by this archive.

The supplied run took approximately 33 seconds for full verification in the execution environment used here; this is a recorded observation, not a hardware-independent performance promise. The uncompressed certificate directory is about 25 MB. Its largest files are sparse coefficient lists, not opaque binary proof objects. SHA-256 hashes in `SHA256SUMS` identify the delivered artifacts.

8.3 Why the natural induction has not yet closed

The data suggest a stronger all-dimensional assertion:

Conjecture 8.1 (Coefficientwise strengthening). For every $n \geq 3$ and $0 \leq j \leq n$, $\mu_{n,j}(\mathbf{1} + \mathbf{x})$ has nonnegative ordinary coefficients. For $n \geq 4$ its support is the entire set of monomials of degree at most j .

The numerical dimension bound has not been removed. In particular, the positive coefficients seen through dimension ten do not justify an induction unless the recursion is shown to preserve the relevant cone of polynomials.

A specific obstacle is that the power-sum operation does not preserve ordinary coefficientwise nonnegativity. Already

$$\sum_{r=1}^Y r^4 = \frac{Y^5}{5} + \frac{Y^4}{2} + \frac{Y^3}{3} - \frac{Y}{30}$$

has a negative coefficient. The magic transform also contains alternating binomial signs. The actual lattice polynomials exhibit cancellations that make the final shifted magic coefficients positive in the certified range, but an arbitrary nonnegative input polynomial need not do so. A proof that merely calls each summation step “positive” would be invalid.

The attempts here isolate three concrete routes for further work. First, it would suffice to build a positivity-preserving recurrence on the boundary family $b_1 = 1$, using Corollary 7.3 to restore the first parameter. Second, a combinatorial interpretation for the integers $[\mathbf{x}^\alpha]n!\mu_{n,j}(\mathbf{1} + \mathbf{x})$ would explain the full support and remove the dimension cutoff simultaneously. Third, one can seek an invariant stronger than ordinary coefficientwise positivity that is preserved by the

finite-sum kernels together with the triangular transform. The published two-parameter theorem remains a useful exact specialization against which any candidate recurrence must be checked.

8.4 Status of the mathematical claims

The theorem in dimensions three through ten is presented as a computer-assisted mathematical result with explicit proof dependencies and reproducible exact data. The cubic formulas, the all-dimensional first coefficient, the endpoint classification, and the cube identity have ordinary proofs printed above. The inequality description and slice machinery are credited to their sources; reciprocity and Brändén’s theorem are stated external inputs. Neither global magic positivity nor global coefficientwise positivity is claimed. The draft has not been independently refereed or checked in a proof assistant, and priority for the consequences derived here has not been established independently.

A Random selection and the full area catalogue

The generator `code/select_area.py` first fixes the ordered list below, obtains a seed using `secrets.randbits(128)`, and calls `random.Random(seed).randrange(120)`. The recorded seed is

89054884759964478034823739394527898187.

It returns zero-based index 32, or one-based entry 33. To replay the same draw without obtaining new entropy:

```
python code/select_area.py --seed 89054884759964478034823739394527898187
```

The selection is reproducible, but its purpose is diversification rather than a mathematical claim about a canonical probability distribution on research areas. In particular, the list itself is an editorial choice.

- | | |
|--|--|
| <ol style="list-style-type: none"> 1. Additive combinatorics 2. Algebraic combinatorics 3. Algebraic geometry 4. Algebraic graph theory 5. Algebraic number theory 6. Algebraic topology 7. Analytic combinatorics 8. Analytic number theory 9. Approximation theory 10. Automata theory 11. Banach space theory 12. Braid groups 13. Calculus of variations 14. Cellular automata 15. Coding theory 16. Combinatorial commutative algebra 17. Combinatorial designs 18. Combinatorial game theory 19. Combinatorial geometry 20. Combinatorics on words | <ol style="list-style-type: none"> 21. Commutative algebra 22. Complex dynamics 23. Computability theory 24. Computational geometry 25. Convex geometry 26. Cryptographic Boolean functions 27. Descriptive set theory 28. Diophantine approximation 29. Difference equations 30. Differential algebra 31. Differential geometry 32. Discrete dynamical systems 33. Discrete geometry 34. Discrepancy theory 35. Distance geometry 36. Enumerative combinatorics 37. Ergodic theory 38. Extremal graph theory 39. Finite fields 40. Finite group theory |
|--|--|

41. Formal languages and grammars
42. Fourier analysis
43. Functional equations
44. General topology
45. Geometric group theory
46. Geometry of numbers
47. Graph colorings
48. Graph polynomials
49. Harmonic analysis
50. Hypergraph theory
51. Incidence geometry
52. Information theory
53. Integer partitions
54. Integral transforms
55. Invariant theory
56. Knot theory
57. Lattice theory
58. Lie algebras
59. Linear algebra and matrix inequalities
60. Logic and proof theory
61. Low-dimensional topology
62. Mathematical billiards
63. Mathematical probability
64. Matroid theory
65. Metric geometry
66. Model theory
67. Modular forms
68. Moment problems
69. Nonassociative algebra
70. Noncommutative algebra
71. Numerical analysis
72. Operator theory
73. Optimization
74. Ordered algebraic structures
75. Order theory
76. Orthogonal polynomials
77. Packing and covering
78. Partial differential equations
79. Permutation patterns
80. Polynomial inequalities
81. Polyhedral combinatorics
82. Potential theory
83. q-Series and basic hypergeometric functions
84. Quiver representations
85. Ramsey theory
86. Random graphs
87. Real algebraic geometry
88. Recurrence sequences
89. Representation theory
90. Riordan arrays
91. Semigroup theory
92. Set theory
93. Spectral graph theory
94. Special functions
95. Substitution dynamical systems
96. Symbolic dynamics
97. Symmetric functions
98. Tensor decomposition
99. Topological dynamics
100. Tropical geometry
101. Type theory
102. Ultrametric analysis
103. Uniform distribution
104. Universal algebra
105. Variational inequalities
106. Vertex algebras
107. Well-quasi-orders
108. Zero-sum theory
109. Discrete differential geometry
110. Algebraic statistics
111. Numerical semigroups
112. Positivity and total positivity
113. Rewriting systems
114. Computational number theory
115. Fractional calculus
116. Nonstandard analysis
117. Geometric measure theory
118. Boolean lattices and set systems
119. Random walks
120. Higher category theory

B Certificate format and audit

A lattice certificate named `lattice_nn.json` stores `dimension`, `scale`, and a lexicographically sorted `terms` list. Each term is `[[a1,...,an],c]` and denotes $c \prod_i b_i^{a_i}$ in the scaled polynomial $n!L_n$. A file `magic_nn_jj.json` uses exactly the same representation but denotes $n!\mu_{n,j}(\mathbf{1} + \mathbf{x})$ in the x_i variables. The distinction between the two variable conventions is essential.

For example, the dimension-three first magic certificate is the list of terms

$$((0, 0, 0), 9), \quad ((0, 0, 1), 11), \quad ((0, 1, 0), 16), \quad ((1, 0, 0), 18),$$

with scale 6. Dividing by six reproduces (4.1). In the dimension-three final certificate, the constant term is absent; its value is exactly zero, not a discarded small floating-point value.

A minimal coefficient audit, independent of any symbolic package, is:

```
import json
from math import comb, factorial
from pathlib import Path

root = Path("data/certificates")
for n in range(3, 11):
    for j in range(n + 1):
        d = json.loads((root / f"magic_{n:02d}_{j:02d}.json").read_text())
        assert d["dimension"] == n and d["scale"] == factorial(n)
        terms = d["terms"]
        exponents = [tuple(a) for a, c in terms]
        assert len(set(exponents)) == len(exponents)
        assert all(len(a) == n and min(a) >= 0 for a in exponents)
        assert all(sum(a) <= j for a in exponents)
        assert all(isinstance(c, int) and c > 0 for a, c in terms)
        assert len(terms) == comb(n + j, j) - ((n, j) == (3, 3))
```

This small audit establishes positivity of the supplied polynomial lists, but by itself does not establish that they are the right polynomials. For that, use the full verifier, which checks their exact construction from the geometric recurrence. Keeping these two obligations separate is part of the proof, not merely a software convention.

References

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